

Conditional statement

- and = both condition must be true then return True
- or = at least one condition is true then return True
- Not = reverse the condition

```
In [1]: i = 0
        while i<5:
            print("Hard work")
            i+=1
```

Hard work
Hard work
Hard work
Hard work
Hard work

```
In [2]: i = 0

        while True:
            if i>5:
                break
            print(i)
            i+=1
```

0
1
2
3
4
5

```
In [3]: i = 0
        while i<10:
            i+=1
            if i==3:
                continue
            print(i)
```

1
2
4
5
6
7
8
9
10

```
In [4]: x = 0
        while x<=5:
            print(x)
            x+=1
        else:
            print("code is over")
```

0
1
2
3
4
5
code is over

```
In [5]: name = "yasir"
        i = 0
        while i < len(name):
            print(name[i])
            i += 1
```

y
a
s
i
r

```
In [6]: # Use a while loop to calculate the sum of numbers from 1 to 50.
```

```
i = 1

result = 0

while i <= 50:
    result += i
    i += 1

print(result)
```

1275

```
In [7]: import time

        count = 5
        while count > 0:
            print(count)
            time.sleep(1)
            count -= 1
        print("Time's up!")
```

5
4
3
2
1
Time's up!

```
In [ ]: # Write a while loop that takes input from the user until they enter 'exi
```

```
while True:
    user_input = input("enter your pasward : ")
    if user_input.lower == 'yasir':
        print('exit the loop')
        break
    else:
        print("enter again : ", user_input)
```

In [9]: *# Reverse a string using a while loop.*

```
name = "yasir arafat"

reverse_name = ""

index = len(name)-1

while index >=0:
    reverse_name += name[index]
    index -= 1

print(reverse_name)
```

tafara risay

In [10]: *# Count the number of vowels in a string using a while loop.*

```
name = "Yasir arafat"

vowel = "aeiou"

number_of_vowel = 0

name_in_lowercase = name.lower()

index = 0

while index < len(name):
    if name_in_lowercase[index] in vowel:
        number_of_vowel += 1
    index +=1

print(number_of_vowel)
```

5

In [11]: *# Print all digits in a string using a while loop.*

```
name = 'y74asir123123araf54at'

index = 0

while index < len(name):
    if name[index].isdigit():
        print(name[index], end=" ")
    index +=1
```

7 4 1 2 3 1 2 3 5 4

In [12]: *# Print all digits in a string using a while loop.*

```
#method 1
num = 6

factorial = 1

while num > 0:
    factorial *= num
    num -= 1
```

```

print(factorial)

#method 2
number = 5

factorial_number = 1

i = 1

while i <= number:
    factorial_number *= i
    i += 1

print(factorial_number)

```

720

120

In [13]: *# Write a program that asks for a password until the correct one is entered*

```

password = ""

while password != "yasir":
    password = input("enter again : ")
    print(f"your password is correct : {password}")

```

your password is correct : yasir

In [14]: *# Check if a string is a palindrome using a while loop.*

```

name = "level"

start = 0

end = len(name) - 1

is_palindromic = True

while start < end:
    if name[start] != name[end]:
        is_palindromic = False
        break
    start += 1
    end -= 1

if is_palindromic:
    print("This number is palindromic")
else:
    print("This number is not palindromic")

```

This number is palindromic

In []: *# Remove all spaces from a string using a while loop.*

```

name = "yassir arafta"
index = 0
result = ""

```

```

x = len(name)

while index < x:
    if name[index] != " ":
        result += name[index]
        index +=1

print(index)

```

In [15]: *# Implement string compression like "aaabb" → "a3b2" using a while loop.*

```

name = "yyyaaasssiiiirrr"
result = "" #first result is empty
index = 0   #satrting at 0 index

while index < len(name):
    count = 1
    while index + 1 < len(name) and name[index] == name[index + 1]:
        count +=1
        index +=1
    result += name[index] + str(count)
    index +=1
print(result)

```

y3a3s3i3r3

In [16]: *# Print the Fibonacci series up to n terms using a while loop.*

```

num = 10

a , b = 0 , 1

count = 0

while count < num:
    print(a, end=" ")
    a , b = b , a + b
    count +=1

```

0 1 1 2 3 5 8 13 21 34

In [17]: *i = 0*

```

while i <= 3:
    j = 0
    while j <=3:
        print(f"i={i},j={j}")
        j+=1
    i+=1

```

```
i=0,j=0
i=0,j=1
i=0,j=2
i=0,j=3
i=1,j=0
i=1,j=1
i=1,j=2
i=1,j=3
i=2,j=0
i=2,j=1
i=2,j=2
i=2,j=3
i=3,j=0
i=3,j=1
i=3,j=2
i=3,j=3
```

In [18]: *#1. Print Even Numbers from 2 to 20*

```
i = 2
while i <= 20:
    print(i)
    i += 2
```

```
2
4
6
8
10
12
14
16
18
20
```

In [19]: `i = 2`

```
while True:
    if i % 2 == 0:
        print(i)
        if i <= 20:
            break

    i += 1
```

```
2
```

In []: